BioLM-NET: an interpretable deep learning model combining prior biological knowledge and contextual LLM gene embeddings on multi-omics data to predict disease

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GitHub Package

Introduction

Biologically informed deep neural networks that connect input to hidden layers based on gene–pathway relationships have gained popularity, but most methods overlook protein–protein (PPI) and protein–DNA interactions (PDI). We introduce **BioLM-NET**, a deep learning framework that integrates single-cell or bulk gene expression and DNA methylation data with prior biological knowledge, including PPI and PDI. BioLM-NET further aggregates latent omics representations at the pathway level via an attention-based pathway layer. We evaluated BioLM-NET on scTrioseq2 colorectal cancer data to predict primary vs. metastatic cells, on TCGA-BRCA, TCGA-GBM, and TCGA-COAD for **cancer subtyping**, and on ROSMAP for **Alzheimer's disease prediction**. BioLM-NET significantly outperformed baselines and state-of-the-art models (P-NET[1], PASNet[2]) on scTrioseq2, TCGA-COAD, and ROSMAP, and was on par with SVM and dense neural networks on TCGA-BRCA. Ablation studies confirmed the significance of PPI, PDI, and the pathway attention layer. Model interpretation further revealed that top features were significantly enriched in GO terms and KEGG pathways, suggesting potential biomarkers and therapeutic targets.

Our Contributions

- **Integrated biological priors**: BioLM-NET jointly incorporates pathways, PPI, and PDI networks within a single architecture.
- **Multi-omics and context-specific design**: Fuses single-cell gene expression, DNA methylation, and large language model (LLM)-derived gene embeddings to capture richer, context-aware biological signals.
- **Robust benchmarking and performance**: Achieves state-of-the-art (SOTA) results across five datasets (cancer and Alzheimer's disease).
- **Enhanced interpretability**: Important features reveal key molecular features driving cancer progression, metastasis, and disease stratification.
- BioLM=NET source code and datasets are available at **https://github.com/bozdaglab/BioLM-NET**

Methodology

- BioLM-NET model architecture is shown in **Fig. 1**. The key components of BioLM-NET are described below.
- **Dual-branch design:** To integrate gene expression and DNA methylation data.
- **Biological priors via PDI, PPI & Pathway mask:** PDI (binary connections linking TFs to their target genes), PPI (connectivity between proteins using STRING database), and pathway (mapping of genes to KEGG pathways) links were incorporated via custom mask matrices.
- **Attention-based pathway layer** – GenePT[3] embeddings integrate gene signals into pathway embeddings by assigning context-aware weights to each gene–pathway link.
- **Fusion module** – concatenates pathway embeddings from both branches and projected through fully connected layers for final prediction.
- **Customized loss function** – Weighted sparse categorical cross-entropy balances classes by assigning higher weights to underrepresented classes.

Model Architecture

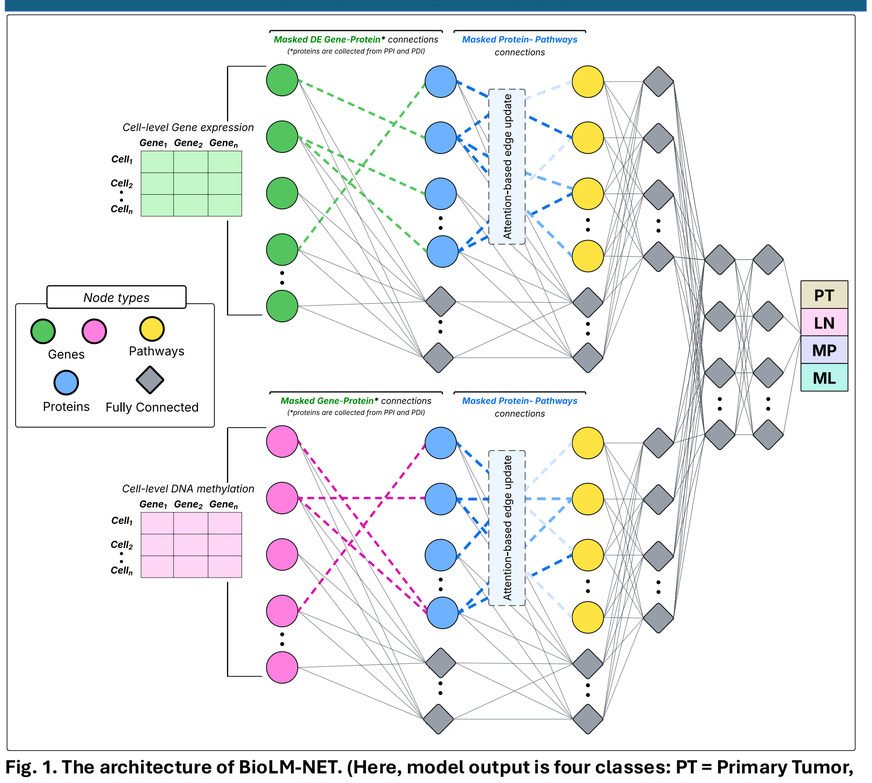

Fig. 1. The architecture of BioLM-NET. (Here, model output is four classes: PT = Primary Tumor, LN = Lymph Node Metastasis, MP= Post-Treatment Liver Metastasis, ML= Liver Metastasis)

Datasets & Preprocessing

- **Datasets:** BioLM-NET was evaluated on five datasets, namely scTrio-seq2, TCGA (BRCA, GBM, COAD), and ROSMAP (**Table 1**).
- **Biological Priors**: PDI (DoRothEA[4], levels A–C) and PPI (STRING[5], score >0.7, top 10% based on the combined score)and **Pathways** were obtained via KEGG pathway enrichment (BH adj. p-value < 0.05).
- **LLM embeddings:** Gene embeddings were computed using pre-trained GenePT model.

Dataset	# samples	# features in GE	# features in DM	# Pathways (GE)	# Pathways (DM)
scTrioseq2	609	1,673	1,712	161	173
BRCA	875	999	999	133	142
GBM	244	500	500	140	102
COAD	256	500	500	103	97
ROSMAP	351	200	200	134	134

Table 1: Overview of the five datasets. GE: Gene Expression data, DM: DNA Methylation data

Results

Comparing BioLM-NET with baseline and SOTA models

- We benchmarked against three ML classifiers (SVM, RF, XGBoost), one Dense NN, and two SOTA biologically informed models (P-NET, PASNet).
- BioLM-NET outperformed all methods on scTrioseq2, COAD, ROSMAP, and tied with Dense NN and SVM on BRCA but performance dropped on GBM due to small sample size and class imbalance (**Fig. 2**).

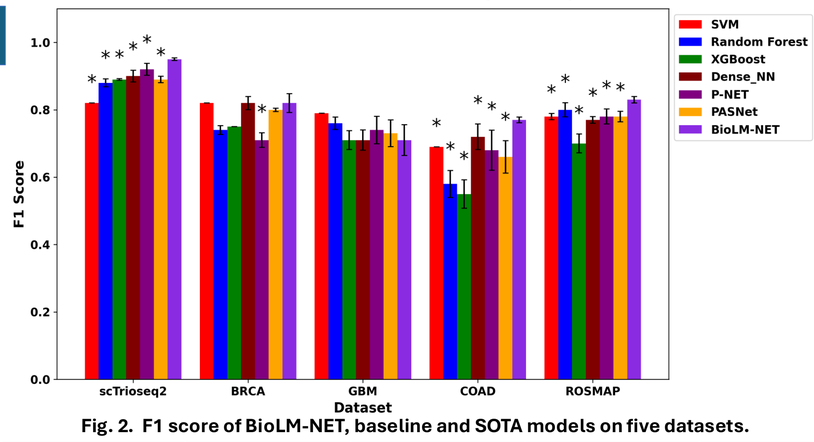

Fig. 2. F1 score of BioLM-NET, baseline and SOTA models on five datasets.

Model	Precision	Recall	F1 Score	Accuracy
GE_Dense	0.90 ± 0.0135	0.90 ± 0.0116	0.89 ± 0.0140	0.89 ± 0.0136
GE_PDI	0.93 ± 0.0113	0.92 ± 0.0104	0.92 ± 0.0105	0.92 ± 0.0096
GE_PPI	0.89 ± 0.0658	0.89 ± 0.0600	0.88 ± 0.0792	0.88 ± 0.0647
GE_PDI_PPI	0.86 ± 0.0896	0.86 ± 0.0605	0.84 ± 0.0783	0.84 ± 0.0580
DM_Dense	0.92 ± 0.0043	0.92 ± 0.002	0.92 ± 0.0035	0.92 ± 0.0040
DM_PDI	0.93 ± 0.0087	0.92 ± 0.0098	0.92 ± 0.0095	0.93 ± 0.0075
DM_PPI	0.91 ± 0.0059	0.91 ± 0.0089	0.91 ± 0.0079	0.92 ± 0.0074
DM_PDI_PPI	0.93 ± 0.0161	0.93 ± 0.0243	0.92 ± 0.0206	0.94 ± 0.0151
GE_DM_Dense	0.92 ± 0.0428	0.89 ± 0.0997	0.88 ± 0.0997	0.89 ± 0.0761
GE_DM_PDI	0.91 ± 0.0134	0.91 ± 0.0096	0.90 ± 0.0124	0.90 ± 0.0120
GE_DM_PPI	0.92 ± 0.0240	0.91 ± 0.0316	0.91 ± 0.0319	0.92 ± 0.0267
GE_DM_PDI_PPI	0.94 ± 0.0153	0.94 ± 0.0113	0.94 ± 0.0126	0.94 ± 0.0111
BioLM-NET	**0.95 ± 0.0062**	**0.96 ± 0.0022**	**0.95 ± 0.0039**	**0.95 ± 0.0039**

Table 2. Effect of adding Prior biological knowledge and multi omics data in scTrioseq2 data. Model naming convention: [Single or Multi Data Modalities]_[Knowledge Type/Layer]

Model	Precision	Recall	F1 Score	Accuracy
BioLM-NET (Without LLM Embedding)	0.94 ± 0.0079	0.93 ± 0.0103	0.93 ± 0.0082	0.94 ± 0.0079
BioLM-NET (Random LLM Embedding)	0.93 ± 0.0168	0.94 ± 0.0250	0.93 ± 0.0215	0.93 ± 0.0204
BioLM-NET	**0.95 ± 0.0062**	**0.96 ± 0.0022**	**0.95 ± 0.0039**	**0.95 ± 0.0039**

Table 3. Effect of adding Pathway Attention Layer in scTrioseq2 data

Predictive performance of BioLM-NET under limited data

- **Robustness test:** Across subsampled training sets (10–90%), BioLM-NET consistently outperformed the Dense NN, demonstrating strong robustness even with limited data **(Fig. 3)**.

Ablation Study

- We conducted an ablation study by evaluating single-modality, multi-modality, and by removing or randomizing GenePT embeddings.
- **Single modality:** GE (F1 = 0.89) improved with PDI (+3%) but dropped with PPI (–1%) or PDI+PPI (–5%); DM performed better (F1 = 0.92) with minimal effect from priors (**Table 2**).
- **Multi-modality:** GE+DM underperformed (F1 = 0.88) but improved with priors (+2–6%) and best results from BioLM-NET with pathway attention **(Table 2).**
- **Attention layer effect**: Removing or randomizing attention scores reduced F1 by ~2%, confirming that true pathway structure and LLM-derived embeddings are essential for prediction **(Table 3).**

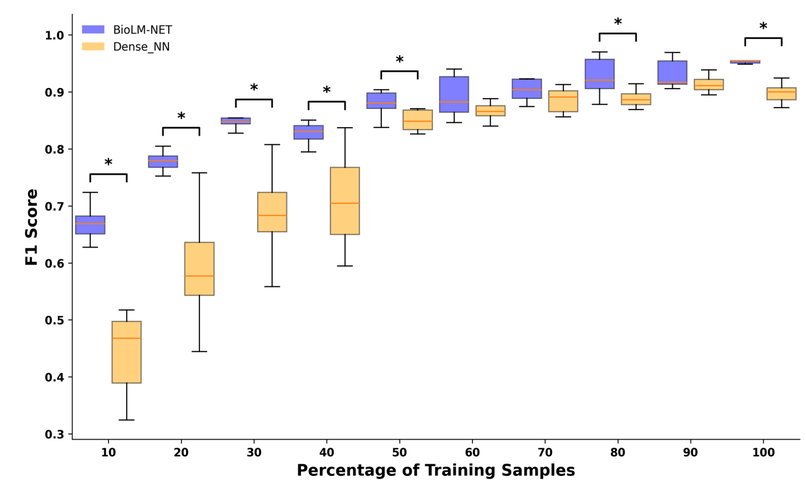

Fig. 3: Boxplots of F1 score for BioLM-NET (blue) and Dense_NN (orange). Wilcoxon rank sum test was performed by comparing BioLM-NET with other models (p-value < 0.05 is denoted by *)

Interpretability of BioLM-NET

- To interpret BioLM-NET, we performed survival analysis using the Kaplan–Meier method on GBM, BRCA, and COAD datasets.
- Top SHAP-selected genes from the gene expression and DNA methylation branches were significantly associated with patient survival, consistent with evidence from prior studies **(Table 4, Fig. 4)**

Dataset	Branch of BioLM-NET	Significant Survival-Related Genes
GBM	Gene Expression	*CBX7, CENPJ, DACH1, NUDT2, PM20D2, RALGAPA1P1, TBX3, WDPCP*
	DNA Methylation	*CPS1, C16orf89, CANT1, CDKN2D, ELSPBP1, GON4L, LDLRAP1*
BRCA	Gene Expression	*BCL2, CXCL1, EIF2S2, FGD3, KDM4B, NXNL2, SOX11, SPARCL1, SUSD3*
	DNA Methylation	*IGFALS, IGFBP4, SCUBE2, SERPINA1*
COAD	Gene Expression	*ASPHD2, COX11, NMNAT1*
	DNA Methylation	*ARRDC1, CAMTA1, CDC14A, DNAJC17, FABP4, FGF22*

Table 4. Significant survival related genes from top 50 SHAP features in GBM, BRCA, and COAD

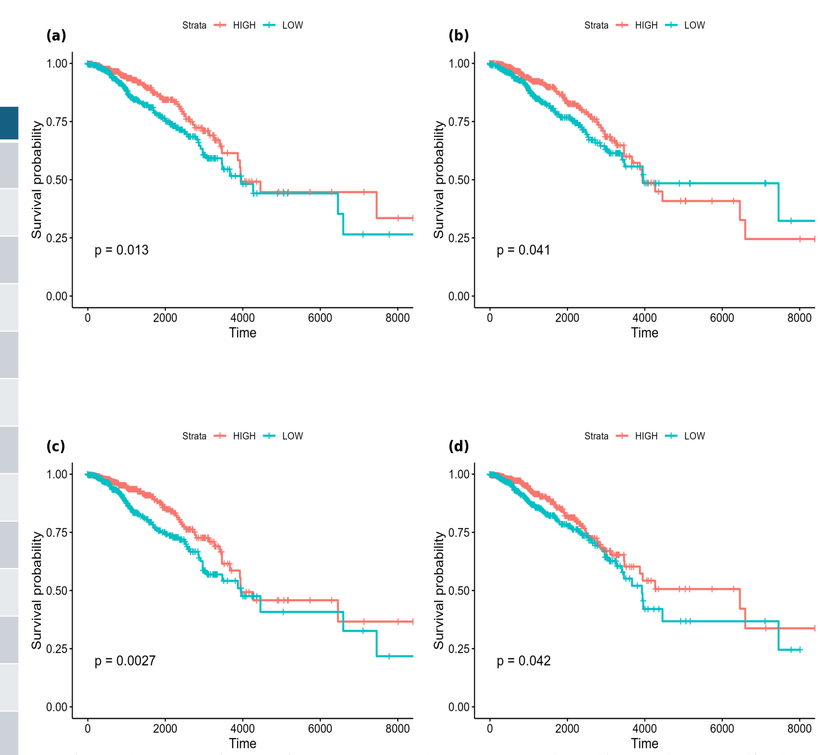

Fig. 4: Kaplan-Meier survival curves for BRCA (DNA Methylation features) patients stratified by high vs. low expression of (a) *IGFALS*, (b) *IGFBP4*, (c) *SCUBE2*, and (d) *SERPINA12*.

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6. The code is available at https://github.com/bozdaglab/BioLM-NET.